



Solutions

COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

Quiz 3 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A
Date:	9 th December, 2025	Maximum Marks:	10	Time:	10 minutes
Student's Name:		Registration Number:			

Question 1: [CLO2-PLO3-C5] [10 Marks]

Use the Implication Table method to optimize the Sequential Logic by reducing the number of states for the FSM shown in Figure 1.

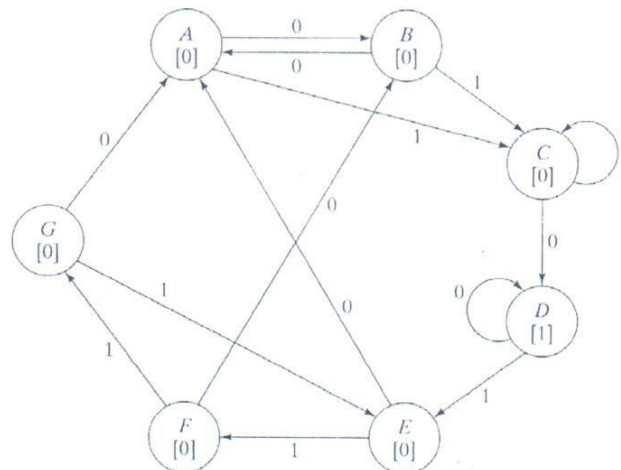


Figure 1

B	A/A					
C	X/B	X/D				
D	X	X	X			
E	B/A	A/A	D/A	X		
F	A/B	A/G	D/E	X	A/B	
G	B/A	A/A	D/A	X	A/A	B/A
	C/E	C/E	C/E		F/E	F/E
	A	B	C	D	E	F

$A=B$
 $E=F$
 $F=G$
 $G=E$
 $E=F=G$

